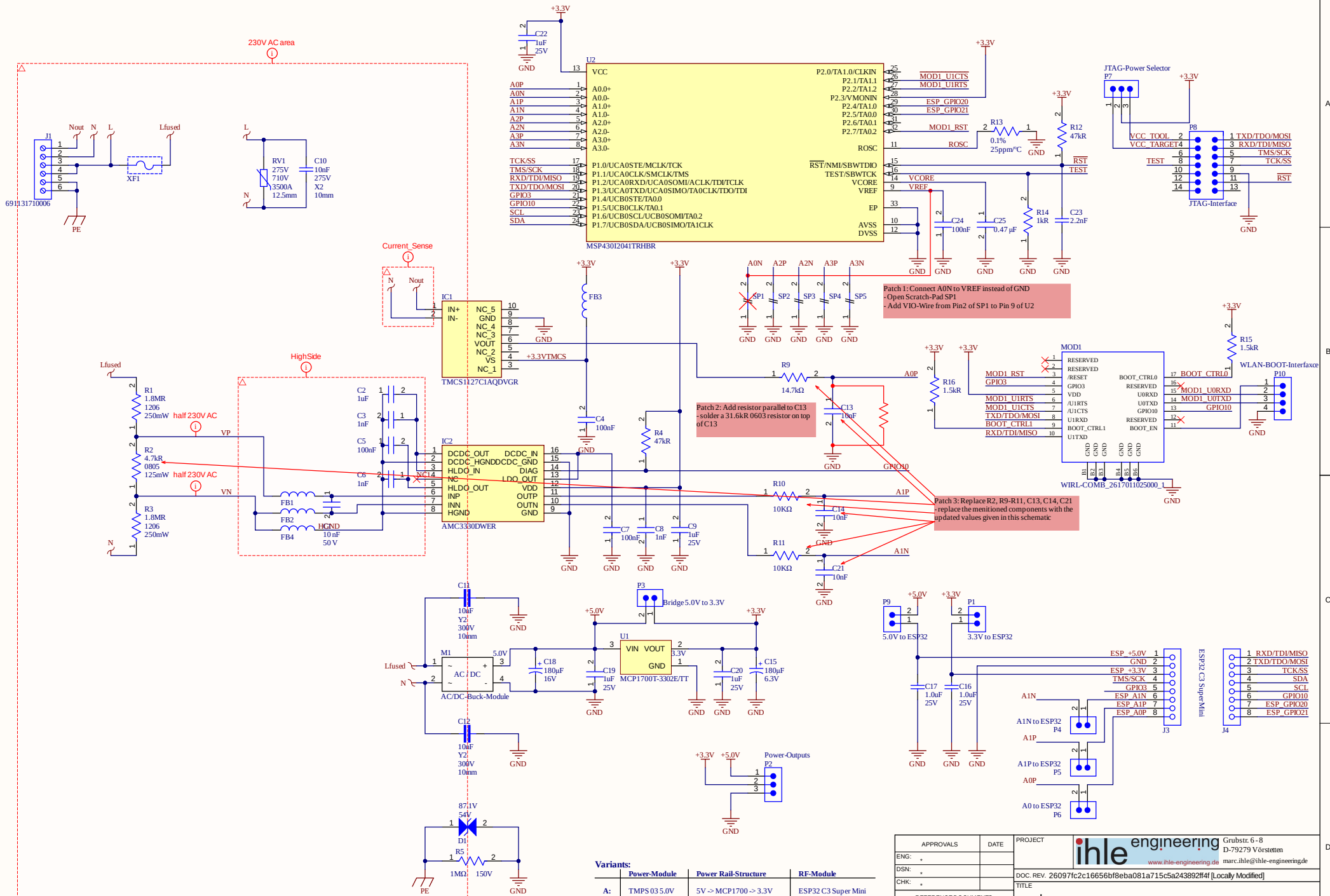




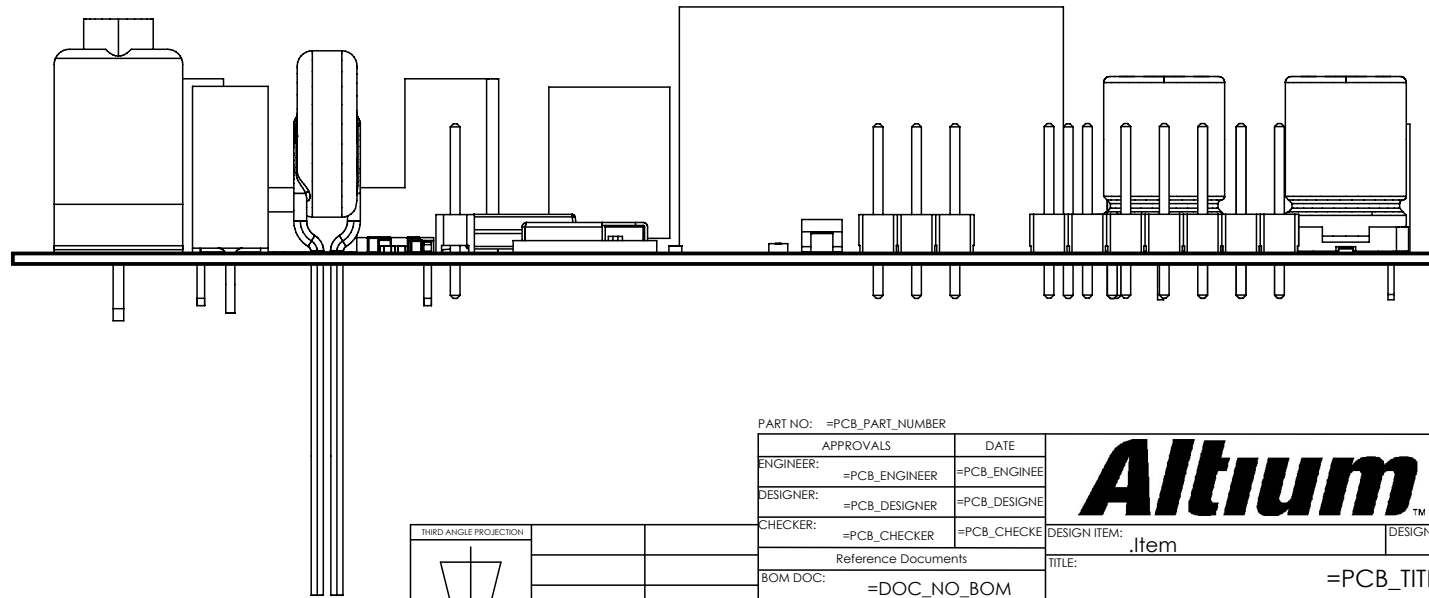
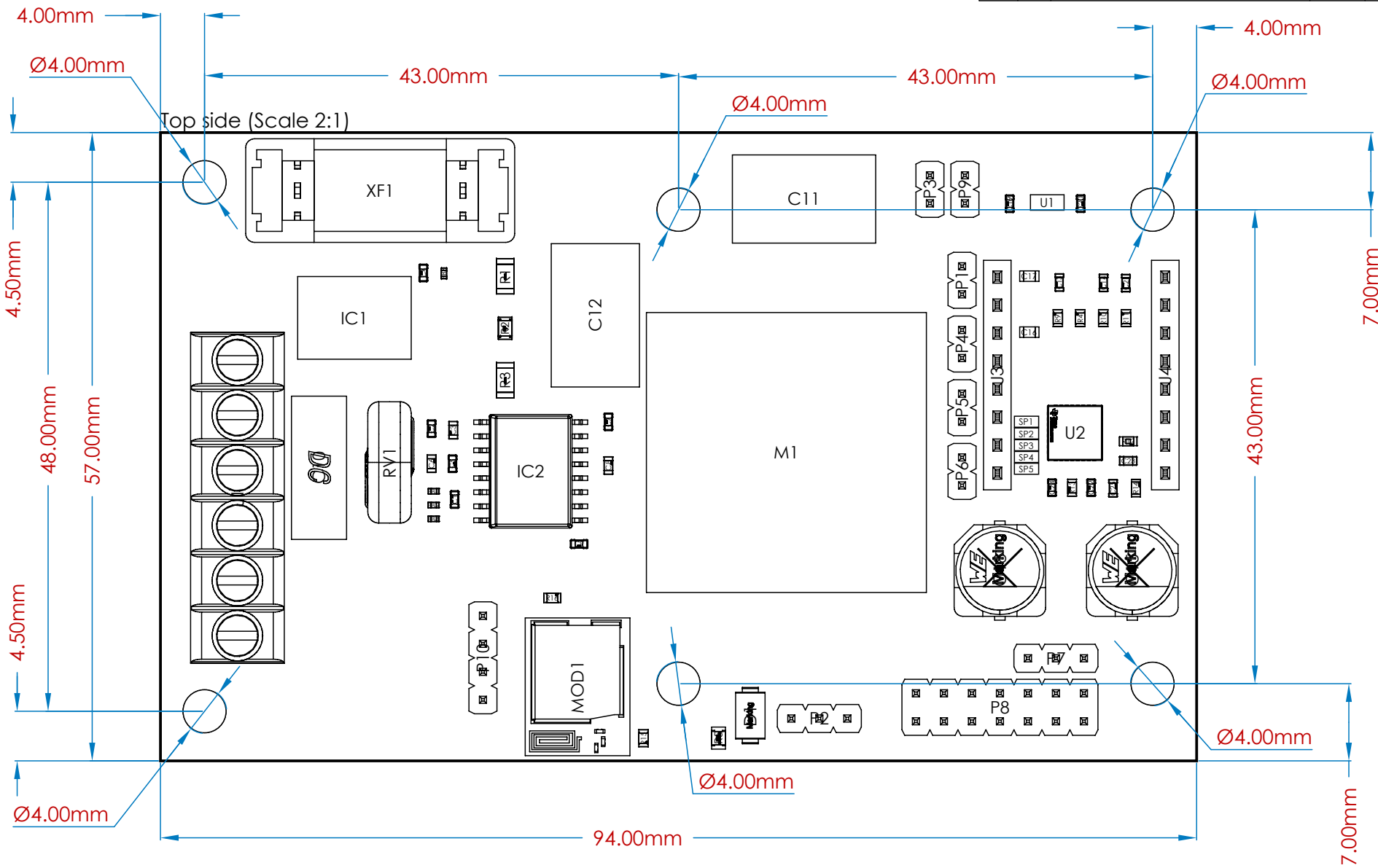
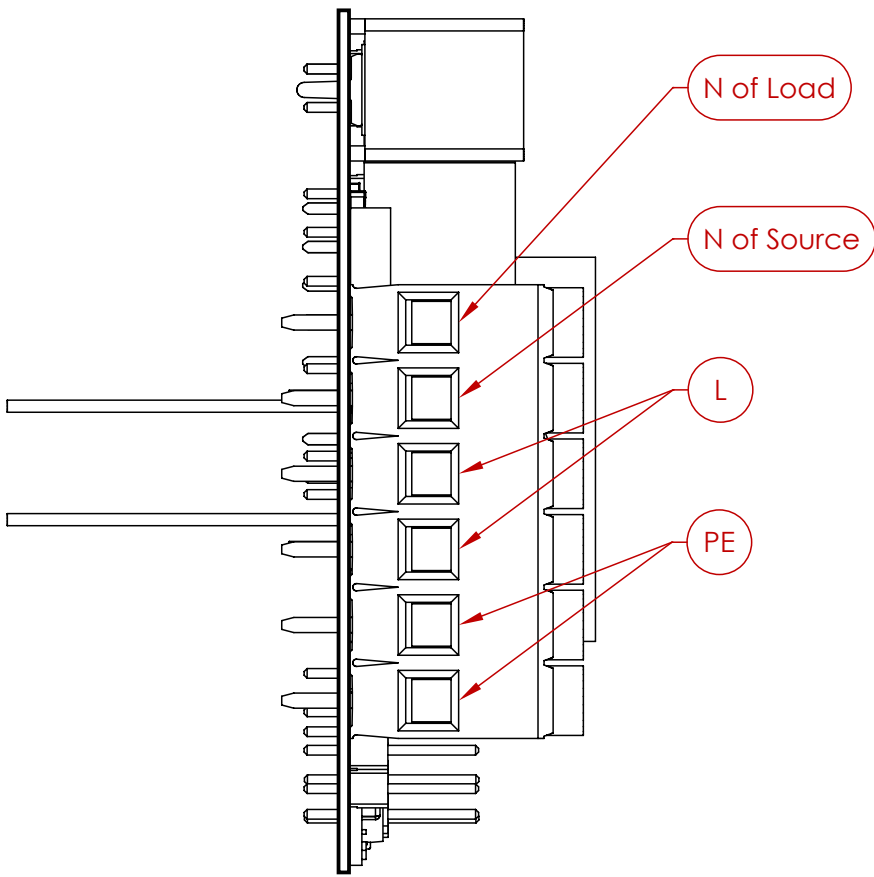
Varianis:

	Power-Module	Power Rail-Structure	RF-Module
A:	TMPS03 5.0V	5V → MCP1700 → 3.3V	ESP32 C3 Super Mini
B:	AM11-12W05V	5V → MCP1700 → 3.3V	ESP32 C3 Super Mini
C:	TMPS03 3.3V	5V = 3.3V (P3 set)	Stephano-I

APPROVALS	DATE	PROJECT	ihle engineering	Grubstr. 6-8 D-79279 Vörsstetten
ENG: *		DOC. REV.	26097fc2c16656bf8eba081a715c5a243892ff4f [Locally Modified]	marc.ihle@ihle-engineering.de
DSN: *		TITLE	★	
CHK: *		REFERENCE DOCUMENTS		
BOM:		SIZE	DWG NO.	REV
ASSY DWG:		A3		
FAB DWG:		FILE NAME	AC-Power-Monitor.SchDoc	SHEET 1 OF 1
PCB DWG:				

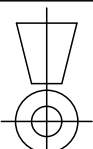
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REV STATUS OF SHEETS	REV	-	-	-	-	-	-	-	-	-	REVISIONS				
	SHEET	-	-	-	-	-	-	-	-	-	ZONE	REV	DESCRIPTION	DATE	APPROVED
											--	-	-----	--/--/--	-, -----
											--	-	-----	--/--/--	-, -----
											--	-	-----	--/--/--	-, -----
											--	-	-----	--/--/--	-, -----
											--	-	-----	--/--/--	-, -----



NOTES UNLESS OTHERWISE SPCEIFIED:

1. Board shape optimized for BOPLA SE 435 DE - 9016 housing
2. NOTE
3. NOTE

PART NO: =PCB_PART_NUMBER		APPROVALS		DATE			
ENGINEER: =PCB_ENGINEER							
DESIGNER: =PCB_DESIGNER							
CHECKER: =PCB_CHECKER							
THIRD ANGLE PROJECTION				DESIGN ITEM: .Item		DESIGN ITEM REVISION: .ItemRevision	
		Reference Documents		TITLE: =PCB_TITLE_1			
		BOM DOC: =DOC_NO_BOM		=DRAF =PCB_TITLE_2			
		ASSY DOC: =DOC_NO_FAB_DWG		SIZE: CAGE CODE: DWG NO: REV:			
		SCH DOC: =DOC_NO_SCH_DWG		A3 =CAGE_CO		=DOC_NO_ASSY_DWG	
		PCB DOC: =PCB_DWG_NO		SCALE: New		AC-Power-Monitor.PCBDwf	
NEXT ASSY USED ON APPLICATION				SHEET: 1 OF 5			

Bill Of Materials

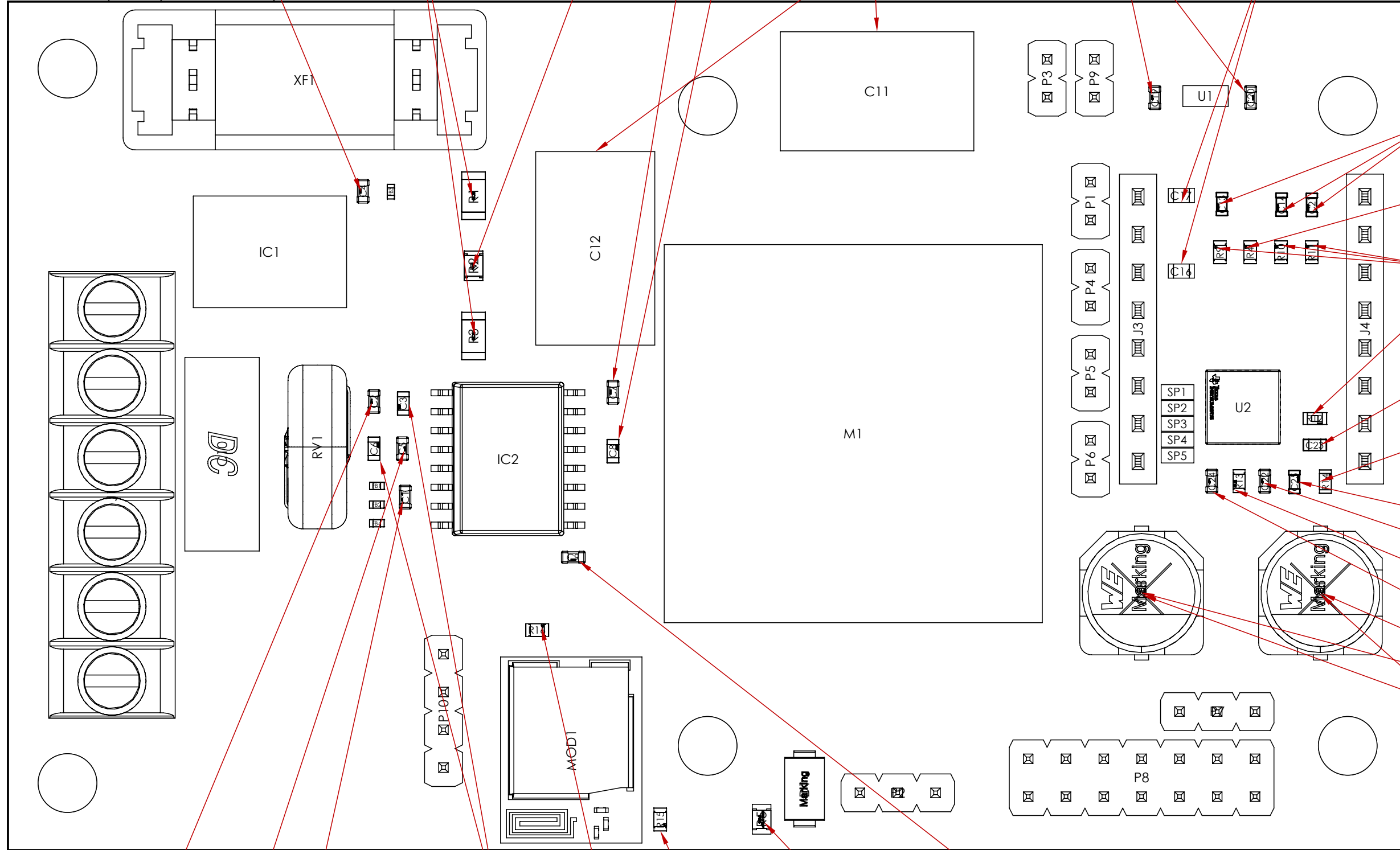
Line #	Designator	Name	Quantity	Purpose
	C1	CC0603KRX7R9BB103	1	
	C2, C9, C19, C20, C22	CGA3E1X7R1E105K080AC	5	
	C3, C6, C8	06035C102KAT2A	3	
	C4, C5, C7, C24	CGA3E2X7R1H104K080AA	4	
	C10	10nF	1	
	C11, C12	MKY22W21003I00	2	
	C13, C14, C21	06035C103KAT4A	3	
	C15	180µF	1	
	C16, C17	C0603C105K3RACTU	2	
	C18	180µF	1	
	C23	885012206085	1	
	C25	C0603C474K4RACTU	1	
	D1	400W, 54VDC	1	
	FB1, FB2, FB3, FB4	1800Ohm	4	
	IC1	TMCS1127C1AQDVGR	1	
	IC2	AMC3330DWER	1	
	J1	691131710006	1	
	J3, J4	61300811821	2	
	M1	AC/DC-Buck-Module	1	
	MOD1	WIRL-COMB_2617011025000_1	1	
	P1, P3, P4, P5, P6, P9	61300211121	6	3.3V to ESP32, Bridge 5.0V to 3.3V, A1N to ESP32, A1P to ESP32, A0 to ESP32, 5.0V to ESP32
	P2, P7	61300311121	2	Power-Outputs, JTAG- Power Selector
	P8	61301421121	1	JTAG-Interface
	P10	61300411121	1	WLAN-BOOT- Interfaxce
	R1, R3	RK73H2BTTD1804F	2	
	R2	CRCW08054K70FKEA	1	
	R4	CRCW060347K0FKEA	1	
	R5	CRGCQ0805F1M0	1	
	R9	ERA-3AEB1472V	1	
	R10, R11	RN73C1J10KBTG	2	
	R12	KTR03EZPF4702	1	
	R13	CPF0603B20KE1	1	
	R14	AC0603FR-071KL	1	
	R15, R16	CRCW06031K50FKEA	2	
	RV1	275VAC, 350VDC	1	
	U1	MCP1700T-3302E/TT	1	
	U2	MSP430I2041TRHBR	1	
	XF1	696106003002	1	

Layer Stack Legend

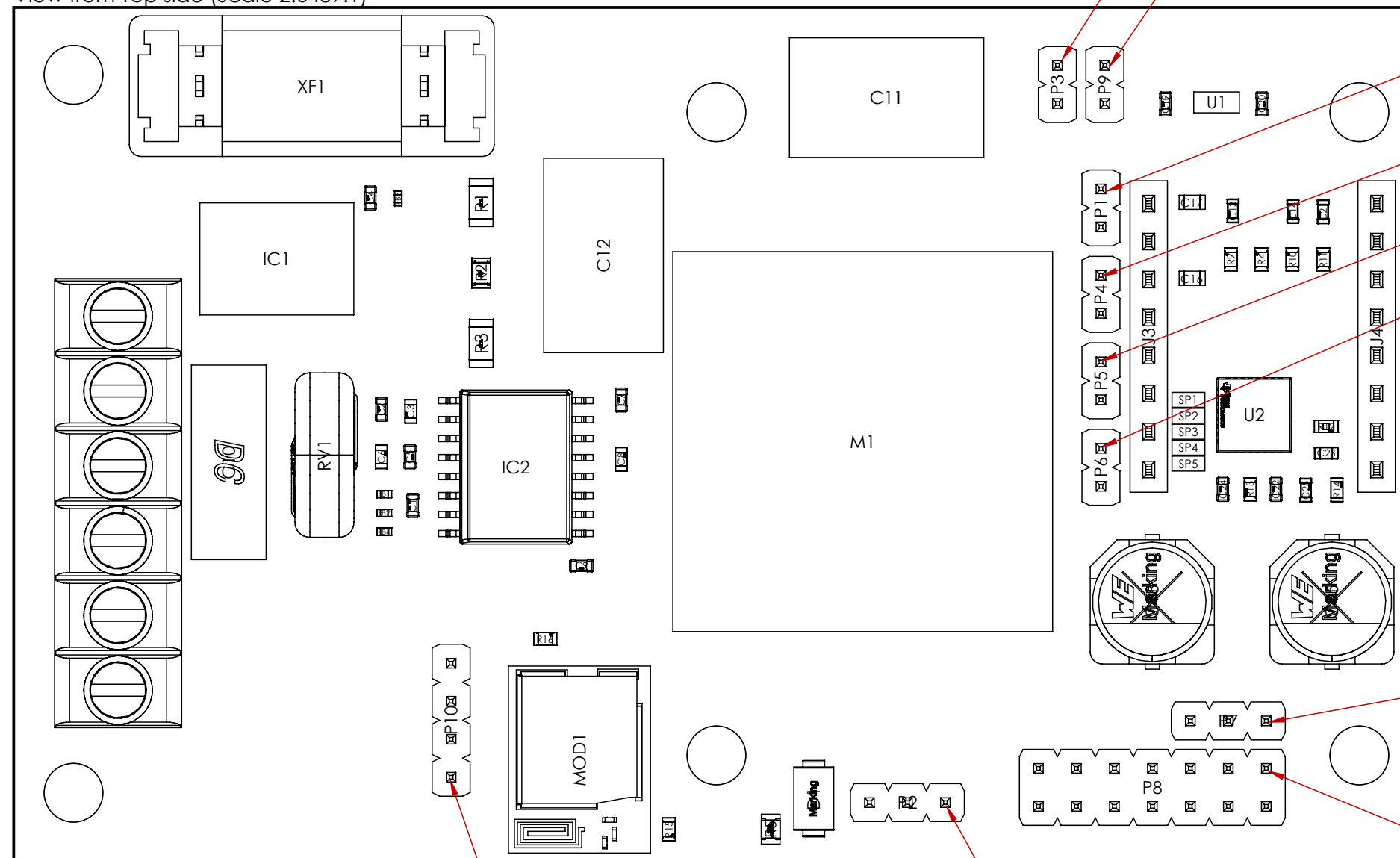
Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.01mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.04mm		Signal	GTL
Prepreg		0.07mm	PP-006	Dielectric	
CF-004	N / GND	0.07mm		Signal	G1
		0.32mm	FR-4	Dielectric	
CF-004	L / VCC	0.07mm		Signal	G2
Prepreg		0.07mm	PP-006	Dielectric	
Copper	Bottom Layer	0.04mm		Signal	GBL
Surface Material	Bottom Solder	0.01mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

Total thickness: 0.69mm

View from Top side (Scale 3.4574:1)



View from Top side (Scale 2.6489:1)



Jumper: Bridge 5.0V-Net to 3.3V-Net
Don't bridge if a 5.0V DC/DC-Converter is in use!

Jumper: Connect 5.0V to ESP32

Jumper: Connect 3.3V to ESP32

Jumper: Connect A1N to ESP32

Jumper: Connect A1P to ESP32

Jumper: Connect A0P to ESP32

Jumper:
right: VCC Tool
left: VCC Target

Pin1, JTAG-Interface

MOD1-Debugger

5.0V, 3.3V, GND



Bill of Materials for AC-Power-Monitor.PrjPcb

Source Data From: AC-Power-Monitor.BomDoc
Variant: None
Report Date: 18.05.2025 11:51

Primary										Secondary			
Altium Vault PN	Designator	Description	Qty	Supplier	Supplier Part Number	Manufacturer	Manufacturer Part Number	Supplier Unit Price	Supplier	Supplier Part Number	Manufacturer	Manufacturer Part Number	Supplier Unit Price
CMP-2100-03600-2	C1	CAP CER 10000PF 50V X7R 0603	1	Mouser	603-CC603KRX7R9B8103	Yageo	CC603KRX7R9B8103	0,0957					
CMP-08246-001507-1	C2, C22	Cap Ceramic 1uF 25V X7R 10% Pad SMD 0603 +125°C Automotive T/R	2	Mouser	810-CGA3E1X7R1E105K	TDK	CGA3E1X7R1E105K080AC	0,1531					
CMP-2006-04378-2	C3, C6	General Purpose Ceramic Capacitor, 0603, 1nF, 10%, X7R, 15%, 50V	2	Mouser	581-06035C102K	Kyocera AVX	06035C102KAT2A	0,0201					
CMP-2000-07546-2	C4, C7	Cap Ceramic 100nF 50V X7R 10% Pad SMD 0603 +125°C Automotive T/R	2	Mouser	810-CGA3E2X7R1H104K	TDK	CGA3E2X7R1H104K080AA	0,0957					
CMP-2000-07546-2	C5	Cap Ceramic 100nF 50V X7R 10% Pad SMD 0603 +125°C Automotive T/R	1	Mouser	810-CGA3E2X7R1H104K	TDK	CGA3E2X7R1H104K080AA	0,0957					
CMP-2006-04378-2	C8	General Purpose Ceramic Capacitor, 0603, 1nF, 10%, X7R, 15%, 50V	1	Mouser	581-06035C102K	Kyocera AVX	06035C102KAT2A	0,0201					
CMP-08246-001507-1	C9, C19, C20	Cap Ceramic 1uF 25V X7R 10% Pad SMD 0603 +125°C Automotive T/R	3	Mouser	810-CGA3E1X7R1E105K	TDK	CGA3E1X7R1E105K080AC	0,1531					
890324023006CS	C10	WCAP-FTX2 Film Capacitor, THT, L13 x W5 x H1.1mm, 10nF, 275V	1	Würth	890324023006CS	Würth	890324023006CS	0,3891					
C, unipolar	C11, C12		2	Mouser	505-MKY22W2100300KS	WIMA	MKY22W2100300KSSD	0,7943					
CMP-2006-02076-2	C13, C14, C21	CAP CER 10000PF 50V X7R 0603	3										
875075155001, 875075355001	C15, C18	WCAP-PSLC AluminumPolymer Capacitor, V-Chip SMT, D8 x H1.17mm, 180µF, 6.3V, WCAP-PSLC AluminumPolymer Capacitor, V-Chip SMT, D8 x H1.17mm, 180µF, 16V	2	NoParanç, Digkey	NoParanç, 732-6488-1-ND	Würth Electronics	875075155001, 875075355001						
CMP-2006-03713-1	C16, C17		2	Mouser	80-C0603C105K3R	KEMET	C0603C105K3RACTU	0,2392					
CMP-03913-53914794-1	C23	885012206085 Multilayer Ceramic Capacitor, Wcap-Cap Series, 2200 pF, 10%, X7R, 50 V, 0603 [1608 Metric] RoHS Compliant: Yes.	1	Mouser	710-885012206085	Würth Electronics	885012206085	0,0957					
CMP-2000-07546-2	C24	Cap Ceramic 100nF 50V X7R 10% Pad SMD 0603 +125°C Automotive T/R	1	Mouser	810-CGA3E2X7R1H104K	TDK	CGA3E2X7R1H104K080AA	0,0957					
CMP-2006-03235-2	C25	CAP CER 0.47UF 16V X7R 0603	1	Mouser	80-C0603C474K4R	KEMET	C0603C474K4RACTU	0,1148					
824501541	D1	WE-TVSP SMT Power TVS Diode, DO-214AC, Bidirectional, 400W, 54VDC	1	Mouser	710-824501541	Würth Electronics	824501541	0,2775					
74269244182	FB1, FB2, FB3, FB4	WE-TKSB Tiny Multilayer Suppression Bead, size 0402, 18000hm, 0.21A	4	Digkey	732-74269244182CT-ND	Würth Electronics	74269244182	0,1053					
TMCS1127C1AQDVGR	IC1	Integrated Circuit	1			Texas Instruments	TMCS1127C1AQDVGR						
AMC3330DWER	IC2	Integrated Circuit	1	Mouser	595-AMC3330DWER	Texas Instruments	AMC3330DWER	6,1900					
691131710006	J1	WR-TBL Series 1317 Horizontal Entry, THT, pitch 5mm, 06p	1	Mouser	710-691131710006	Würth Electronics	691131710006	3,9100					
61300811821	J3, J4	WR-PHD Socket Header, THT, Vertical, pitch 2.54mm, 1 Row, 8P	2	Mouser	710-61300811821	Würth Electronics	61300811821	0,4019					
ACDC-Buck-Module	M1		1	Arrow Electronics	TMPS 03-105	Traco Power	TMPS03-105	16,6500					
WRL-COMB_2617011025000_1	MOD1	Stephano-1 - Combined 2.4 GHz Wi-Fi and Bluetooth® LE 5.0 Module	1	Digkey	732-2617011025000DKR-ND	Würth Elektronik	2617011025000	10,1400					
61300211121	P1, P3, P4, P5, P6, P9	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 2P	6	Mouser	710-61300211121	Würth Electronics	61300211121	0,1244					
61300311121	P2, P7	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 3P	2	Mouser	710-61300311121	Würth Electronics	61300311121	0,1244					
61301421121	P8	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 2 Row, 14P	1	Mouser	710-61301421121	Würth Electronics	61301421121	0,9187					
61300411121	P10	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 4P	1	Mouser	710-61300411121	Würth Electronics	61300411121	0,1818					
CMP-2003-01283-2	R1, R3	1.5kΩ ±1% 0.25W Chip Resistor 1206 Thick Film AEC-Q200 Qualified	2	Mouser	660-FK73H2BT1D1804F	KOA Speer	FK73H2BT1D1804F	0,0957					
CMP-2100-03615-2	R2	RES Thick Film, 4.7kΩ, 1%, 0.125W, 100ppm/°C, 0805	1										
CMP-2000-03379-3	R4	RES SMD 47k OHM 1% 1/10W 0603	1	Mouser	71-CRCW0603-47K-E3	Vishay	CRCW060347K0FKEA	0,0957					
CMP-03211-002463-1	R5	CRGQ0805 1MΩ 0.125W ±100ppm/°C 1% 150V	1	Newark	73AC2562	TE Connectivity	CRGQ0805F1M0	0,0038					
CMP-2000-00913-2	R9	RES SMD 14.7kOHM 0.1% 1/10W 0603	1										
CMP-03211-002042-1	R10, R11	Thin Film Resistors RN 0603 10KΩ 0.1% 10ppm/°C	2										
CMP-08839-030818-1	R12	Res Thick Film 0603 47kΩ 1% 0.1W ±100ppm/°C Automotive Paper T/R	1	Mouser	755-KTR03E2PF4702	Rohm	KTR03E2PF4702	0,1531					
CMP-03211-006634-1	R13	RES SMD 20KΩ 0.1% 1/16W 0603	1	Arrow Electronics	CPF603B20KE1	TE Connectivity	CPF603B20KE1	0,1772					
CMP-2000-04628-3	R14	RES 1kΩ 1.00% 0.1W 0603	1	Digkey	311-1KL0DKR-ND	Yageo	A0603PR-071AL	0,0957					
CMP-2000-02995-2	R15, R16	RES Thick Film, 1.5kΩ, 1%, 0.1W, 100ppm/°C, 0603	2	Newark	61M5331	Vishay	CRCW6031K50FKEA	0,0172					
820412711	RV1	WE-VD Disk Varistor, THT, 275VAC, 350VDC, 190µF	1	Mouser	710-820412711	Würth Electronics	820412711	0,5455					
CMP-0136-00788-1	UI1	Low Quiescent Current LDO, 3-Pin SOT-23, Extended Temperature, Tape and Reel	1	Arrow Electronics	MCPL700T-3002EY1T	Microchip	MCPL700T-3002EY1T	0,4267					
CMP-04869-000074-1	U2	EC MCU 16BIT 32KB FLASH 32VCFN	1	Mouser	595-MSPA3020411T9-BR	Texas Instruments	MSPA3020411T9-BR	2,9800					
696106003002	XF1	WR-FSH Fuseholder THT Blocks PCB Clip Cover	1	Mouser	710-696106003002	Würth Electronics	696106003002	0,5550					